



Linear cellular automata-based impulse noise identification and filtration of degraded images

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Abstract

In this paper, four impulse noise filters based on modifications of linear cellular automata (LCA) are proposed and evaluated. Each of these filters make use of an adaptive neighborhood to provide efficient noise filtration at varying noise densities. The LCA model works asynchronously and makes the proposed filters computationally efficient. Peak signal to noise ratio and structural similarity (SSIM) index metrics are used to provide objective analysis of the proposed filters.

Keywords Salt-and-pepper noise · Impulse noise · Cellular automata · Structural similarity index · Peak signal to noise ratio

1 Introduction

Digital images have become an effective medium for communicating different types of information concerning a wide variety of areas including industrial processes, health-care, social networking, remote sensing, and education. Use of smart phones and other portable devices with cameras has made capturing of digital images easier than ever [2]. Noise generally refers to unwanted data that distort and hide the original information carried in digital images. In extreme cases, noise change or distort the semantics of information carried in digital images to an extent that human or computer vision systems fail to recognize the original image content. Therefore, it becomes immensely important to filter out this noise from the images. The presence of noise in digital images can affect the appropriateness of results produced by computer vision systems [20].

Most of the impulse noise filters proposed in literature tend to perform poor as the noise densities increase in the digital images. Lack of adaptability to varying levels of noise have paved way for the application of cellular automata (CA) to noise filtration. In the last decade, various CA-based impulse noise filters have been proposed. Information about the type

of noise in digital images plays a significant role in the subsequent image de-noising process [4]. The presence of noise can be detected based on appearance of the image, and hence, impulse noise is measured based on the brightness quality [12]. Most of the detection algorithms are based on the assumption that noise free images consist of locally smoothly varying areas separated by edges and a noise pixel takes a gray value substantially larger or smaller than those of its neighbors. The median-based impulse noise detectors and filters are not able to make distinction between fine lines and impulses and as such result in removal of such fine lines. Most linear filters like the average filter (AF) are not effective for filtering impulse noise from digital images [24] because de-noising capabilities of these filters deteriorate considerably and remove original image information in addition to the noise data.

Filters, such as arithmetic filters (AF), median filters (MF), center weighted median filter (CWMF) and weighted median filter (WMF), process all pixels in the noise corrupted image, and, as such, there is some loss of original information because of processing the uncorrupted pixels. Therefore, different filters are proposed that first try to identify the corrupted pixel and process the pixel only if the pixel is labeled corrupted. Switching median filter (SMF) [28] is based on MF that filters identified corrupted pixels only. Adaptive Center weighted Median Filter (ACWMF) [16] was proposed to filter images with small contamination ratio. A rough estimate of the corruption ratio is calculated to compute a central weight in an adaptive manner. The problem with this filter is the introduction of unwanted smoothing artifacts after noise

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filtering [15]. These filters are not able to adapt to the varying densities of noise and thus result in poor filtration at high noise density [9]. As such, many CA-based noise filtration algorithms [13], [3] have been proposed in last few years.

[23] proposed an efficient fuzzy CA-based impulse noise filter. [24] also proposed a salt and noise filtering method based on fuzzy CA. CA-based noise filter proposed in [27] that works with a fixed boundary, i.e., local transition rule is applied to only non-boundary cells. A factor α in the range $[0, 0.5]$ is used in the CA local transition function. [25] proposed a fuzzy CA-based impulse noise filtration method. [17] proposed a CA-based method to filter out the speckle noise in ultrasound images using non-local mean. [30] proposed a salt-and-pepper noise filtering method using CA under Moore neighborhood configuration N_M with $r = 1$. [20] proposed an efficient CA-based impulse noise filtration method that employ primary CA rules under Moore neighborhood configuration to filter digital images.

[22] proposed a support vector machine (SVM)-based impulse noise filter, and [12] proposed a method for impulse noise identification and removal from grayscale images. Support vector neural network (SVNN)-based fuzzy hybrid filter is used to identify and remove the noise from noise corrupted images. [14] proposed a filtering technique for filtering impulse noise from color images. The filter is switching and based on cumulated distances between processed pixel and its adjacent pixels.

Table 1 lists the advantages and disadvantages of the existing techniques.

Work proposed in this paper uses the linear cellular automata (LCA) model for impulse noise filtration. The proposed filters process the pixels/cells asynchronously in order to filter corrupted pixels only. Using LCA model asynchronously makes the model computationally very simple and experimentation with different neighborhood configurations help to provide better filtration even at high noise density. Main challenge faced by standard noise filters like Wiener filter, Median filter is that they fail to achieve acceptable levels of noise filtering when high density noise is present in the image. The problem is that these filters work using statistical estimates of the neighboring pixels and apply such estimates to the noise corrupted pixel. These estimates often include noisy pixel outliers resulting in values far from the actual value of the pixels to be filtered for noise. The proposed method avoids this problem by excluding such outliers while calculating a value for the pixel to be filtered. Objective of the proposed method is to develop a real-time noise filtering method that filters images for high noise densities. The specific contributions of the work are listed as under:

1. A computationally simple impulse noise filtration algorithm based on modification of linear cellular automata model is proposed.

4096 (i-2,j-2)	2048 (i-2,j-1)	1024 (i-2,j)	512 (i-2,j+1)	16777216 (i-2,j+2)
8192 (i-1,j-2)	4 (i-1,j-1)	2 (i-1,j)	256 (i-1,j+1)	8388608 (i-1,j+2)
16384 (i,j-2)	8 (i,j-1)	1 (i,j)	128 (i,j+1)	4194304 (i,j+2)
32768 (i+1,j-2)	16 (i+1,j-1)	32 (i+1,j)	64 (i+1,j+1)	2097152 (i+1,j+2)
65536 (i+2,j-2)	131072 (i+2,j-1)	262144 (i+2,j)	524288 (i+2,j+1)	1048576 (i+2,j+2)

Fig. 1 Linear cellular automata (LCA) primary rule model. A total of 25 primary rules expressible as 2^k , where k is in the interval $[0,25]$

2. The filter is designed to recover both grayscale as well as RGB color images corrupted by high density noise.
3. The proposed filter is computationally very simple involving only comparison and assignment operations and at the same time ensuring superior noise filtration results.

The rest of this paper is organized as follows: section (2) describes the concept of CA and LCA; section (3) presents the proposed noise filtration model; section (4) provides the detailed results and discussion followed by the conclusion and future scope of the study in section (5).

2 Cellular automata

Cellular Automata (CA) are time and space discrete mathematical models described in late 1940s by John von Neumann and Stanislaw Ulam. CA have been considered one of the most pronounced parallel computational tools in the recent era of nature and bio-inspired computing. Because of their local interaction, simplicity of design and inherent parallelism, many hardware implementations of CA have been realized in the literature. Therefore, CA are ideal models that can be applied to many image processing tasks requiring pixel processing in the spatial domain. CA local interactions, simple design and inherent parallelism make them suitable for different digital image processing techniques [1]. CA find application in a wide variety of fields including image scrambling [10], image enhancement [11], image encryption [6] and image forensics [5].

2.1 Linear cellular automata model

Qadir et al. [20] proposed the linear cellular automata (LCA) model for studying various geometrical transformations of digital images. The primary rule map for the LCA model as proposed in the work is given in Fig. 1.

The LCA primary rule model is based upon the binomial theorem and as such there are 24 primary rules and $2^{25} - 1$

Table 1 Advantages and disadvantages of existing impulse noise filtering techniques

Technique	Pros	Cons
Linear filters (Arithmetic/Weiner Filter)	Easy to Implement; conceptually simple	Edge distortion; Image blurring;
Non-Linear filters (Median Filter)	Edge preservation; less noise propagation	Poor filtering performance against high density noise.
Switching Median Filter (SWF) [18]	Split noise treatment; remove relatively higher density impulse noise	Edge distortion; Difficult to choose a suitable threshold.
Adaptive center weighted median filter (ACWMF) [16]	Noisy regions are treated according to the noise density	Corruption ratio estimation not true always
Selvapeter et al. [26]	Easy to understand and implement; computationally efficient	Edge distortion; Propagate noise in local pixel neighborhood;
Jeelani and Qadir [9]	Good high density noise filtering	Vertical and Horizontal noise propagation
Qadir and Shoosha [20]	Computationally efficient	Edge distortion of images with high noise density

secondary rules. The cell marked 1 in Fig. 1 is called the cell under examination. The next state of this cell depends upon the state of cells in its neighborhood as given in eq. (1). If the next state of the cell under examination depends only on one cell, the rules are called primary rules. And, if the next state of the cell depends on more than one neighboring cell, the rules are called secondary rules. The LCA primary rule numbers correspond to the numbers in each cell in Fig. 1. The secondary rule number used to represent and consider multiple cells in the neighborhood of cell with primary rule number 1 can be obtained by adding their corresponding primary rule numbers. Therefore, each primary rule number can be represented as 2^k where $k \in \{x : 0 \leq x \leq 24\}$. The possible secondary rule numbers lie in the range 2^0 to $2^{25} - 1$.

$$N_M(i, j, r) = \{C(i', j') : |i' - i| + |j' - j| \leq r\} \quad (1)$$

Here, C denotes the LCA lattice; $C(i, j)$ represents a cell in LCA at index (i, j) . The LCA model-based techniques are shown to be useful in various image processing applications like, edge detection [19], image segmentation, replication [21], salt and pepper noise reduction [24] and so on. However, LCA primary rules have not been studied extensively for impulsive noise reduction particularly up to the radius equal to 2 in the Moore neighborhood (N_M) setup. So, to study impulsive noise reduction extensively, here we study all the primary rules of this model up to radius equal to 2 using different transition functions.

3 Proposed method

The Linear Cellular Automata (LCA) model proposed in [19] is suitably modified in the proposed method. Specifically, the correspondence between cell locations and primary rule

numbers is modified. Noise corrupted image C is the initial LCA configuration. Periodic boundary condition is used to avoid problem of missing cells at the edges of the LCA configuration.

The correlation between adjacent pixels in an image is high [7]. The strong correlation between adjacent pixels in row, column and diagonal orientation is shown to be extremely high in [6] and [10]. The implication of the results shown in these studies is that the pixel values of adjacent pixels are almost same. The proposed filters make use of this feature and first detect a noise corrupted pixel and systematically try to assign the pixel value of an adjacent noiseless pixel, as close as possible, to the noisy pixel. The basis for the proposed filters is in choosing the specific pixels and the order in which these pixels are assigned to a noise corrupted pixel using different LCA rules.

The proposed filter begins by detecting a noise corrupted pixel by using a fuzzy switching median noise detection technique as proposed in [29]. Once the corrupted noise pixel is detected, the LCA-based noise filtration model is invoked. The proposed noise filtration model is based on the LCA primary rule model given in Fig. 1. The pixel (*cell under examination*) is marked as primary rule 1. The general LCA-based noise filtration model is given in eq. (2). The intensity value of the noise corrupted cell/pixel $C(i, j)_1^t$ at time step t is replaced by the intensity value of a cell $C(i', j')_\eta^t$ identified by primary rule number η at time $t + 1$ to obtain the noise filtered pixel $C(i, j)_1^{t+1}$.

$$C(i, j)_1^{t+1} = \delta(C(i, j)_1^t, C(i', j')_\eta^t) \quad (2)$$

where, δ is the local transition function, $C(i, j)_1^t$ is the current noisy pixel, t is the time step and 1 is the primary rule number corresponding to the location (i, j) , $C(i', j')_\eta^t$ is the uncorrupted neighborhood cell identified by the primary rule

number η with $\eta = 2^k$ and $1 \leq k \leq 24$, the cell at location $(i', j') \in N_M(i, j, 2)$ and t is the time step, and $C(i, j)_1^{t+1}$ is the noise filtered pixel at time step $t + 1$.

The noise filtration model is implemented by considering different primary rules to check the performance of these rules to filter out noise from digital images at varying densities. We start by considering only four neighboring cells of the cell (corrupted pixel) at location (i, j) and identified by primary rule number 1. Subsequently, more cells are considered in the following filters. All the following filters process pixels from bottom to top and from right to left in order to avoid vertical noise propagation as discussed in [9].

3.1 Filter 1

Filter 1 (F1) is based on the general filter design given in eq. (2). F1 considers only four neighborhood cells $(C(i', j')_\eta^t)$, and each cell is identified by the primary rule number η ; where, $\eta = 2^{n_k}$ and $n_k = n_1 + (k - 1)2$, $n_1 = 1$, and $2 \leq k \leq 4$.

$\forall C(i, j)_1^t$ in C corrupted by impulse noise
if $(C(i, j)_{2^{n_k}}^t)$ is a corrupted pixel (for $k = 1$ to 4)
then $k = k + 1$
otherwise, $C(i, j)_1^{t+1} = C(i', j')_{2^{n_k}}^t$ (3)

Therefore, F1 uses primary rules 2, 8, 32 and 128 in this order. The pixel/cell $C(i, j)_1^t$ is tested for impulse noise and if the pixel is corrupted, the intensity value of pixel/cell corresponding to the primary rule number 2^{n_k} is assigned to $C(i, j)_1^t$ using δ . But, if pixel identified by the primary rule number 2^{n_k} is also corrupted by the impulse noise, then the primary rule $2^{n_{k+1}}$ is used as given in eq. (3).

3.2 Filter 2

Filter 2 (F2) is also based on the general filter design given in eq. (2). F2 first considers the same cells as considered by F1. If all these cells under consideration are noise corrupted, then F2 considers cells $(C(i', j')_\eta^t)$ identified by the primary rules η for noise filtration using given in eq. (3).

Here, $\eta = 2^{n_k}$ and $n_k = n_1 + (k - 1)2$, $n_1 = 2$, and $2 \leq k \leq 4$.

Therefore, F2 uses primary rules of F1 first and if all these rules fail, F2 uses primary rules 4, 16, 64 and 256 in this order.

3.3 Filter 3

Filter 3 (F3) first considers the same cells as considered by F2 and in the same order. If all these cells under consideration are noise corrupted, then F3 considers cells $(C(i', j')_\eta^t)$ identified by the primary rules η using eq. (3).

where, $\eta = 2^{n_k}$ and $n_k = n_1 + (k - 1)4$, $n_1 = 10$, and $2 \leq k \leq 4$.

F3 first uses all the primary rules considered by filter F2 and if all the rules fail, then F3 uses primary rules 1024, 16384, 262144 and 4194304 in this order.

3.4 Filter 4

Filter 4 (F4) first considers the same cells as considered by the filter F3 and in the same order. If all these cells under consideration are noise corrupted, then F4 considers cells $(C(i', j')_\eta^t)$ identified by the primary rules η as given in (4). Here, $\eta = 2^{n_k}$ and $n_k = n_1 + (k - 1)2$, $n_1 = 9$, and $2 \leq k \leq 8$.

$\forall C(i, j)_1^t$ in C corrupted by impulse noise
if $(C(i, j)_{2^{n_k}}^t)$ is a corrupted pixel (for $k = 1$ to 8)
then $k = k + 1$
otherwise, $C(i, j)_1^{t+1} = C(i', j')_{2^{n_k}}^t$ (4)

Subsequently, F4 considers remaining four cells identified by primary rule numbers η for noise filtration using eq. (3). where, $\eta = 2^{n_k}$ and $n_k = n_1 + (k - 1)4$, $n_1 = 12$, and $2 \leq k \leq 4$.

F4 uses all the primary rules considered by filter F3 first and if all the rules under consideration fail, then F4 first uses primary rules $2^9, 2^{11}, 2^{13}, 2^{15}, 2^{17}, 2^{19}, 2^{21}$ and 2^{23} . In case of very high noise density, all preceding rules may fail and in which case F4 considers primary rules $2^{12}, 2^{16}, 2^{20}$ and 2^{24} in this order.

4 Results and discussion

The proposed noise filtration model is implemented using Matlab R2016a, and the test images are obtained from the Berkeley Segmentation Dataset and Benchmark ([www.eecs.berkeley.edu/ Research/ Projects/ CS/ vision/ bsds](http://www.eecs.berkeley.edu/Research/Projects/CS/vision/bsds)). Set of 50 test images and 20 color images are selected at random from this dataset and used in different experiments to evaluate the performance of proposed filters. The LCA model proposed in [19] is suitably modified to use the model in the proposed noise filtration method. Specifically, the correspondence between cell locations and primary rule numbers is modified as shown in Fig. 1. The noise corrupted image C is the initial LCA configuration. Periodic boundary condition is used to avoid problem of missing cells at the edges of the LCA configuration.

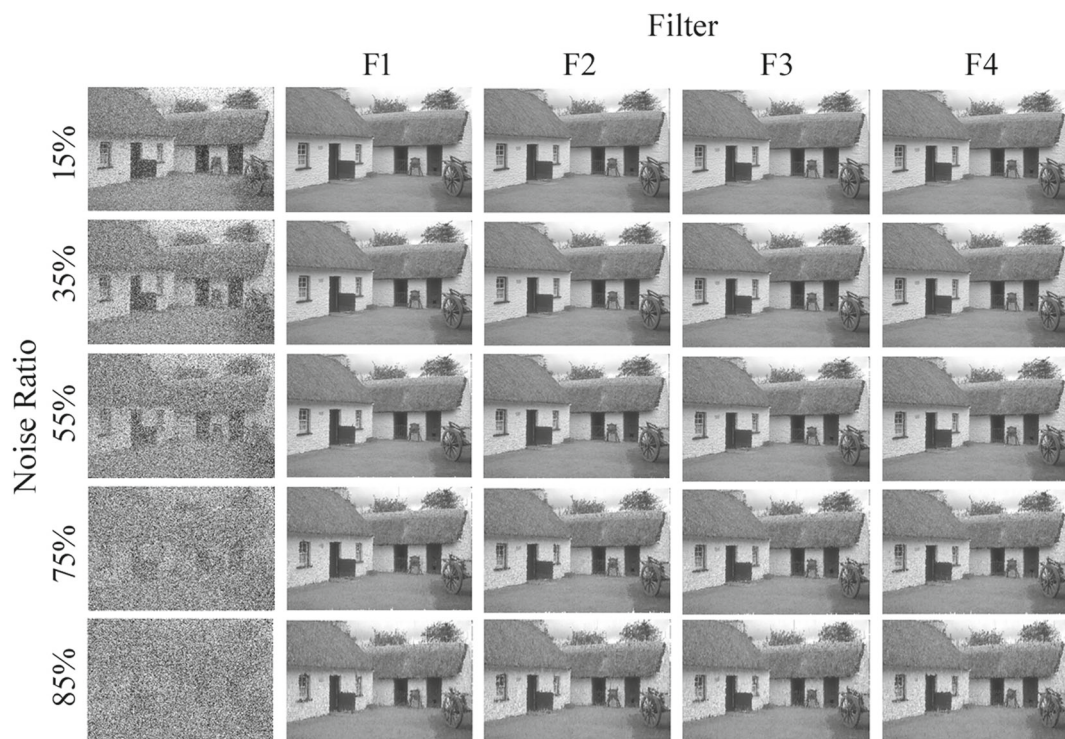


Fig. 2 Filtration results of Test image: **385028.jpg** with noise density ranging between 15 and 85% using proposed filters F1, F2, F3, and F4

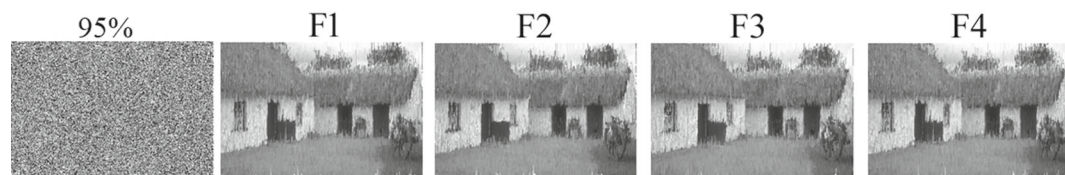


Fig. 3 Filtration results of Test image: **385028.jpg** with noise density=95% using proposed filters F1, F2, F3, and F4

4.1 Comparative analysis of proposed filters

Figure 2 shows subjective results of filtering the 385028.jpg test image using the proposed filters at noise densities ranging between 15 and 85%. It can be seen from the figure that the filtration results are acceptable for all noise densities, but a careful examination of the filtered images shows that the edges get distorted as the noise increase in the input image. Figure 3 shows the filtration results for the same test image at 95% noise density. It can be seen from the figure that the filtered images are corrupted near the bottom. The edges are also distorted with vertical noise propagation.

Proposed filters are computationally simple and need to perform $O(mn)$ operations to noise filter a corrupted image, where $m \times n$ is the size of input image. Basically, the proposed filters look for the uncorrupted pixels in the local neighborhood of the current corrupted pixel so that its intensity value can be assigned to the current corrupted pixel. In the worst case, no neighborhood pixel is uncorrupted and the filter need to check each pixel. All the proposed filters require

$N[(m \times n)\rho\%]$, where N is the number of neighborhood cells considered, $m \times n$ is the input image size and ρ is the noise density. The value of $N=4$ for F1, $N=9$ for F2, $N=12$ for F3, and $N=24$ for F4. Therefore, all the filters have the worst case complexity of $O(mn)$.

4.1.1 PSNR analysis of the proposed method

The filtering performance of the proposed filters is first analyzed using the PSNR metric. Impulse noise ratios ranging between 5% and 95% are used to corrupt 50 test images of different sizes. The proposed filters are used to recover the noise corrupted images, and the PSNR values for these images are recorded. Figure 4a shows a plot of the averaged PSNR values for the same set of test images as used for SSIM analysis. Again, F2 is shown to perform comparatively better filtration in terms of PSNR at high noise density, whereas F1 performs comparatively poor. Filters F2, F3 and F4 perform closely up to 45% noise. Beyond 45% noise, F2 start to show comparatively small PSNR, whereas PSNR values obtained using

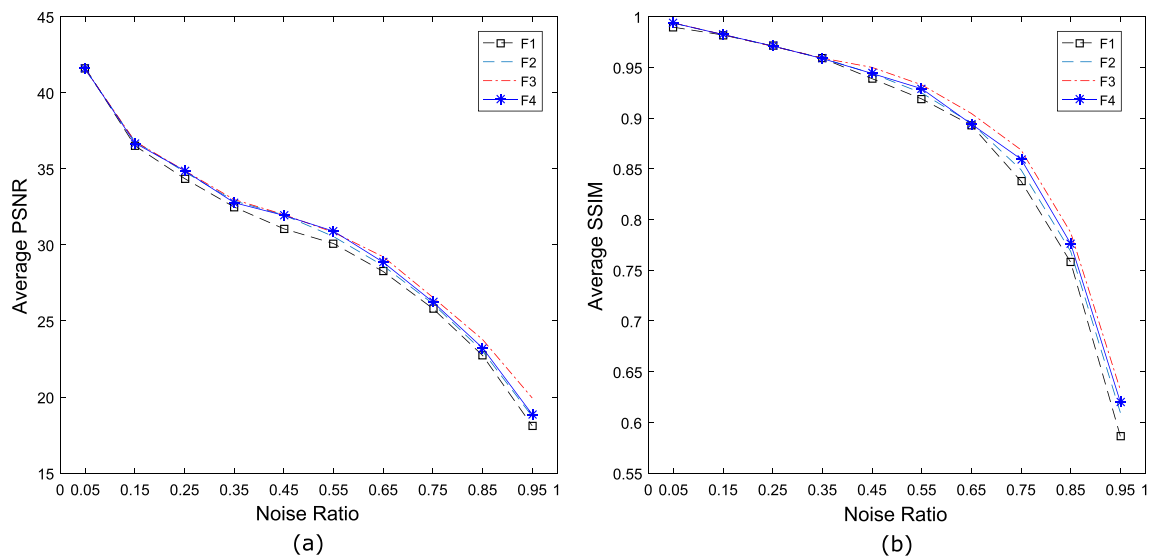


Fig. 4 PSNR and SSIM analysis of proposed filters (F1, F2, F3 and F4). **a** Plot of averaged PSNR values obtained for 50 test images obtained using proposed filters at noise ratios ranging between 5 and 95%, **b** Plot

of averaged SSIM values obtained for 50 test images obtained using proposed filters at noise ratios ranging between 5 and 95%

F3 start to improve slightly and F4 remains somewhere in between F2 and F3. F1 performs slightly bad throughout the noise ratios. We hypothesize that F1 tend to perform comparatively poor because it considers only 4 neighborhood pixels for noise filtration and, thus at high noise density it might not be able to use any neighboring pixel for noise filtration because all the pixels under consideration are also detected to be corrupted by noise. F3 considers 12 neighborhood pixels, whereas F4 take 24 neighborhood pixels under consideration and as the experimental results suggest, it is basically enough to consider just 12 pixels to achieve good filtration results.

4.1.2 SSIM analysis of the proposed method

The SSIM analysis of the proposed filters result in similar behavior as achieved with the PSNR analysis. Figure 4b shows a plot the average SSIM values obtained for the test images. It is evident from the figure that the filtering performance of each filter reduces with increase in noise density. Further, each of the proposed filters shows very similar SSIM behavior, but F3 performs slightly better comparatively, whereas F1 shows comparatively poor SSIM values.

4.2 Comparative analysis of proposed filters with standard and CA-based filters

Figure 5 presents the subjective results of filtering the test image 25098.jpg corrupted with 55% noise density using standard and CA-based filters. Figure 5h is the test image filtered using the proposed (F3) filter. A careful examination

of the figure reveals that the edge details are comparatively more precisely preserved using the proposed filter.

Experimental results in [8] suggest that arithmetic/ mean (AF), median (MF) and switching median filters (SMF) perform better filtration with 25-cell neighborhood compared to 4-cell, 9-cell and 12-cell neighborhoods. In this section, therefore, we consider only 12-cell versions of these filters for comparison purpose. CA-based noise filtration techniques proposed in [26], [20] and [8] are also implemented in Matlab R2016a for comparison purpose. Weiner Filter (WF) and Adaptive Center Weighted Median Filter (ACWMF) are also considered for comparative analysis of the proposed filter (F3). A set of 35 grayscale and 20 color test images are filtered using the filters at varying noise densities and a subjective comparison using SSIM and PSNR with proposed filter (F3) is presented below.

Figure 6a presents averaged PSNR values obtained for the test images using filters as a line plot. Similarly, average SSIM values obtained for the test images using different filters are shown in Figure 6b. It can be seen from the figures that non CA-based filters as well as the CA-based filter proposed in [26] show rapid decrease in metric values as noise density increases beyond 45%. The proposed filter (F3) performs comparatively better than CA-filter proposed in [20] and slightly better than the filter proposed in [8].

4.3 Color image filtration

A set of 20 color images is considered, and each RGB image is divided into three channels and filtered using standard, CA-based and proposed filters. The averaged PSNR and SSIM

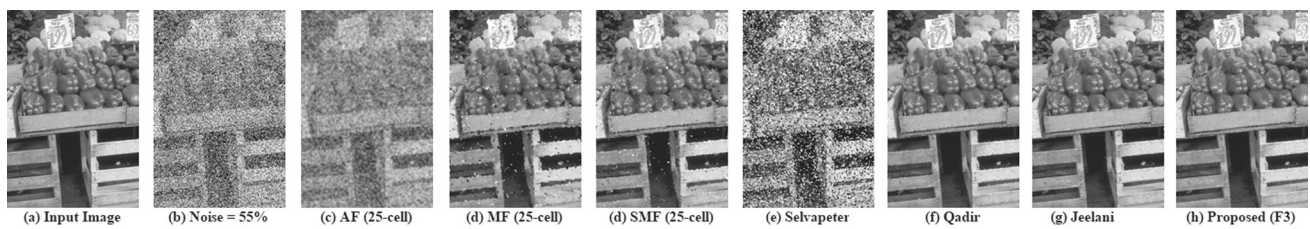


Fig. 5 Filtration results of test image **25098.jpg** corrupted with 55% noise using standard, CA-based filters and proposed filter (F3)

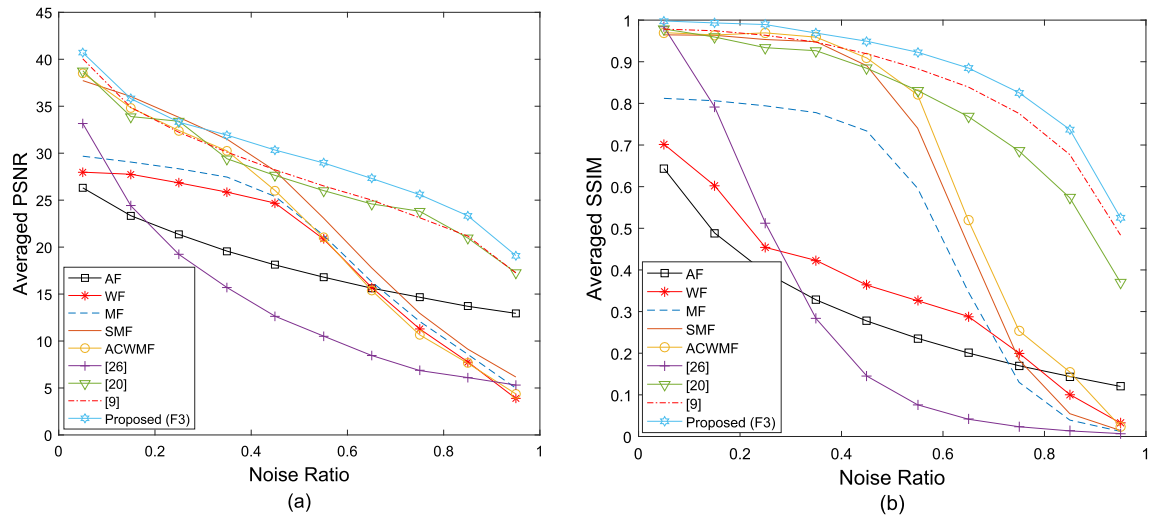


Fig. 6 Comparative PSNR and SSIM analysis of proposed filters (F1, F2, F3 and F4). **a** Plot of averaged PSNR values obtained for 50 test images using proposed and standard/ CA-based filters at noise ratios

ranging between 5 and 95%, **b** Plot of averaged SSIM values obtained for 50 test images using proposed and standard/ CA-based filters at noise ratios ranging between 5 and 95%

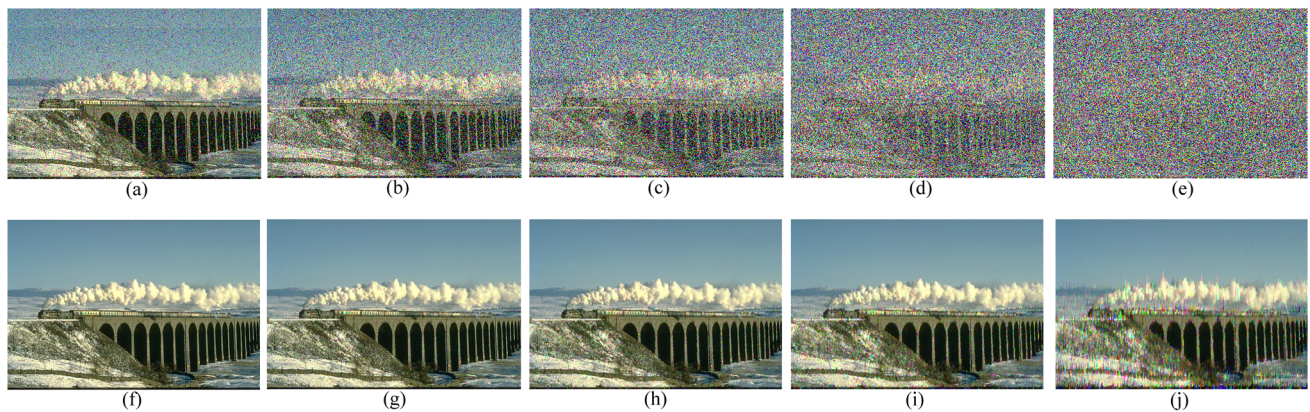


Fig. 7 Filtration results of test image **182053.jpg** using proposed filter F3. Figures **a–e** Noise corrupted images with noise ratio 0.1, 0.3, 0.5, 0.7 and 0.9. Figures **f–j** Corresponding Noise filtered images using filter F3

results are shown in Fig. 6a and Figure 6, respectively. Proposed filters are applied to RGB color images by dividing the RGB image into its three constituent channels. Next, each channel is processed using the filters presented in Section 3. Figure 7 shows the noise filtered images of test image 182053.jpg. Quality of noise filtered image obtained after filtering Fig. 7e at noise ratio 0.9 is comparatively poor. First, the vertical noise propagation near the edges is clearly visible. Second, because of processing each channel independently, the noise propagation in each channel also express itself as red, green and blue regions in the final image obtained after combining the channels.

5 Conclusion

A computationally efficient noise filtration method based on linear cellular automata (LCA) model under different primary rule orders is presented that can filter impulse noise across a wide range of noise densities ranging between 5%–95% effectively but for higher noise densities the vertical noise propagation distort the images adversely affecting the quality of the restored image. Filter 3 (F3) results in best filtration results both subjectively as well as objectively in terms of SSIM and PSNR. The proposed method uses comparison and assignment operations only for noise filtration making it a computationally effective method. Computational efficiency and simplicity are the main advantages of the proposed method. Experimental results of the comparison with many standard as well as CA-based methods suggests a significant improvement in noise filtration process using the proposed method. In future, the authors intend to improve the proposed filter by addressing the problem of edge distortion at high noise density.

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